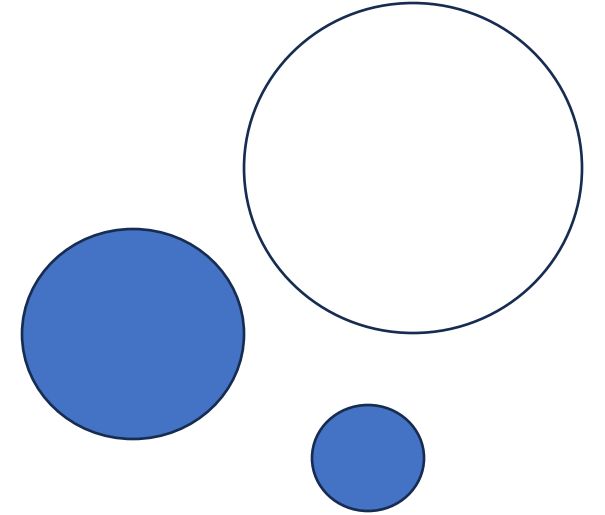




EDWIN SETSOAFIA KWADZO

Mechatronics Engineer | Aspiring Robotics & AI Engineer | Robotics
Research Enthusiast | Robotic Manipulators | Bio-Inspired Robotics |
ML & AI



setsoafiaedwin7@gmail.com



<https://github.com/ayamEdwin>

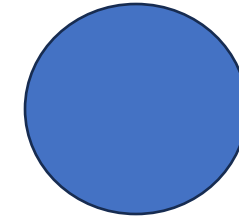


Edwin Setsoafia



ABOUT ME

I am currently a Mechatronics Engineering graduate who is passionate about Robotics and A.I research. I received a Bachelor of Technology in Mechatronics Engineering from Koforidua Technical University, Ghana. My background in mechatronics has enriched my engineering skills in the diverse fields of Mechanical, Electrical/Electronics, Embedded Systems, Robotics and Control Systems.



I believe in a future where Autonomous and Intelligent Robots will one day take on various human-centric roles in industries and homes. To make this a reality, I have developed a deep interest in Intelligent Robots. By combining my mechatronics skills with robot perception, planning and control, I am ready to contribute to a developing world where intelligent robots will not only do the laundry but assist the elderly and also help people to be better at their jobs through collaboration on tasks. This can only be achieved with break-throughs in robotic manipulation research which I seek to contribute to. This portfolio contains projects that I have worked on as well as what I have learned from these projects and from mentoring others.

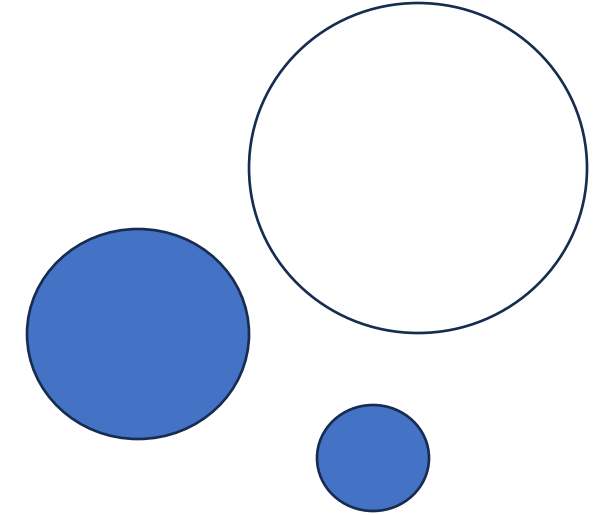


RESEARCH INTERESTS

- ☐ Bio-Inspired Robotics
- ☐ Machine Learning & AI
- ☐ Robotic manipulators
- ☐ Robot Perception

BACKGROUND

- ☐ B. Tech. Mechatronics Engineering, KTU, 2025



CERTIFICATIONS

- ☐ KICAD Phils Lab
- ☐ Skill Lync, Introduction to F.E.A
- ☐ Zero 2 Mastery, Deep Learning with Pytorch

HOBBIES

- ☐ Reading
- ☐ Creative Writing

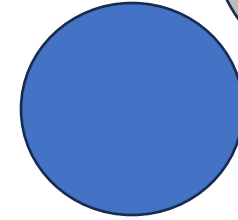


ACADEMIC PROJECTS

- ☐ Bachelor Thesis I & II : Glass Crusher
- ☐ MTR-323 Robotics : Robotron Manipulator
- ☐ Digital Clock
- ☐ mbed - API

INTERNSHIPS

- ☐ Fortress Visitor Log



PERSONAL PROJECTS

- ☐ Custom Arduino Uno R3
- ☐ 5V Linear Power Supply
- ☐ Self-Balancing Robot Simulation (MATLAB)

ONGOING PROJECTS

- ☐ Robotic Hand for Grasping objects – inspired by the dexterity of the human hand
- ☐ DarkLight Documentation and Forum

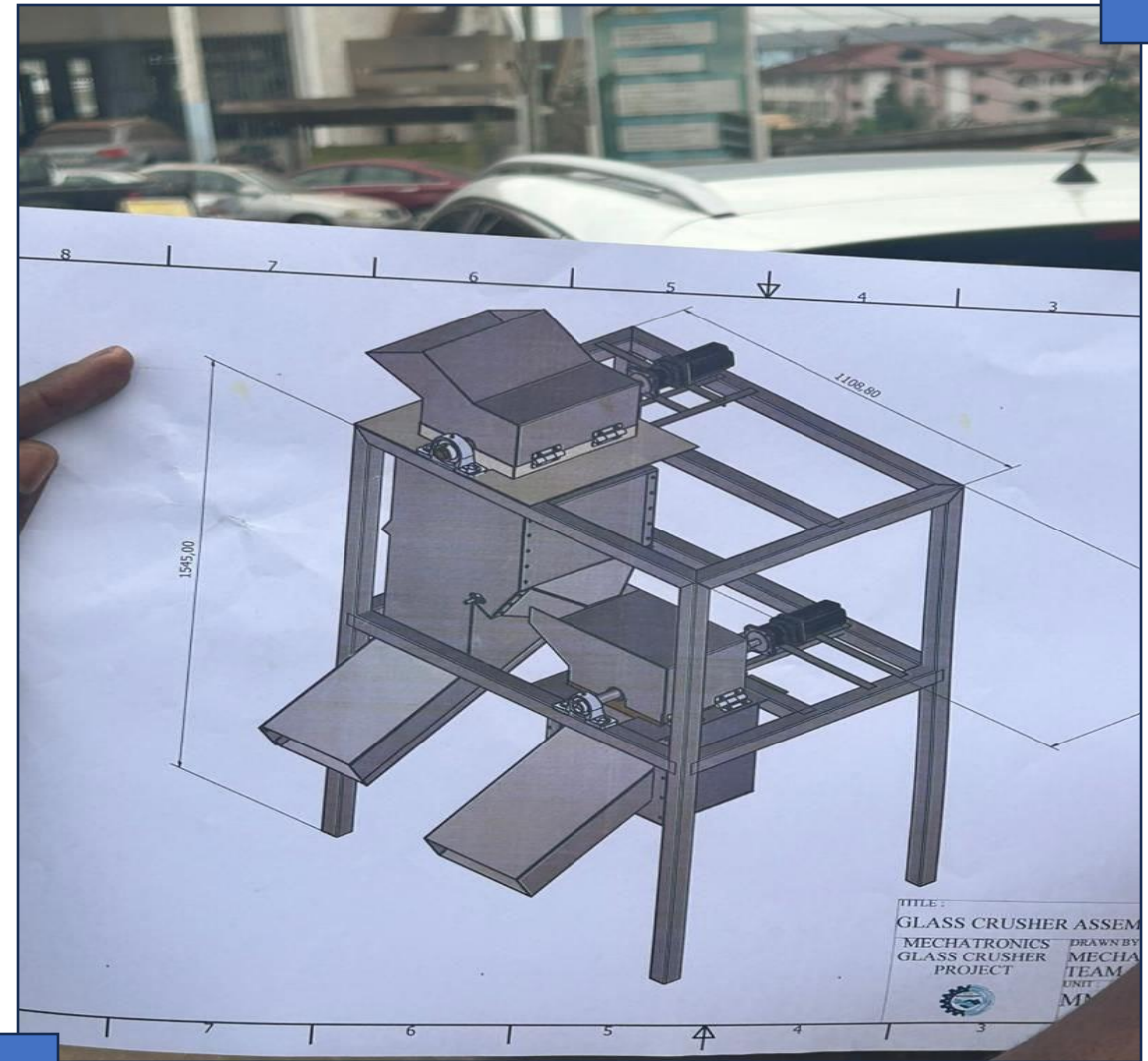
BACHELOR THESIS I,II – GLASS CRUSHER (Year 4)

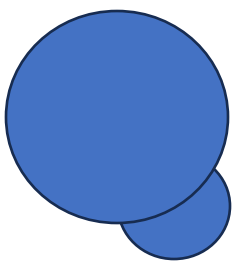
What is MTR-25 Crusher?

MTR-25 Crusher, is the first portable Glass Crusher designed for local manufacturing businesses. It comes with software implemented safety features as well as dual crushing chambers for fine and coarse glass grain output. As part of a team, I designed and implemented a firmware architecture on STM32F0 microcontroller to control the entire operation of the MTR-25 Crusher.

Project Overview

Glass waste accumulates in landfills and poses environmental and safety challenges in Ghana and other urban areas. Our team designed a semi-automated Glass Crusher system to enhance recycling efficiency, incorporate robust motor control, and improve operator safety. This project involved embedded firmware development managing motor control algorithms, gate operations, and user interface tasks for a sustainable, cost-effective crushing solution.

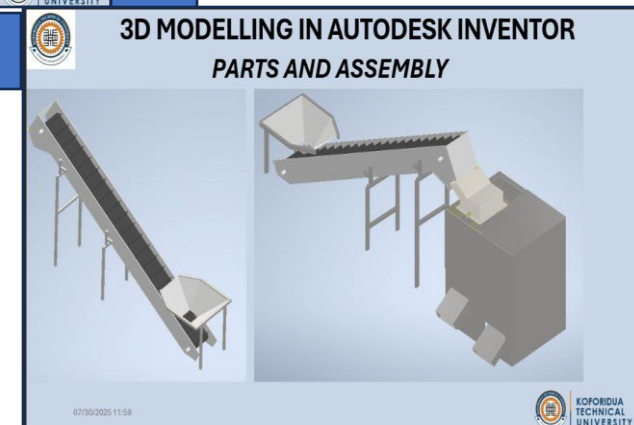
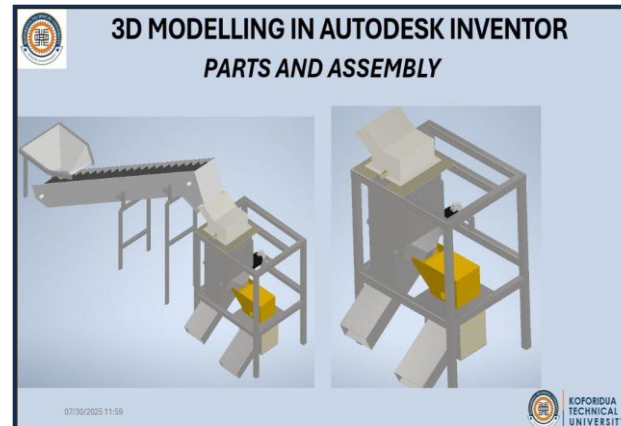




BACHELOR THESIS I,II – GLASS CRUSHER (Year 4)

Motivation

Manual glass crushing is inefficient and hazardous for operators. Large-scale industrial crushers are costly and inaccessible to many local recycling SMEs. We aimed to develop a low-cost, reliable control system for a semi-automated crusher that improves crushing precision and safety, supporting local sustainable waste management and circular economy initiatives.



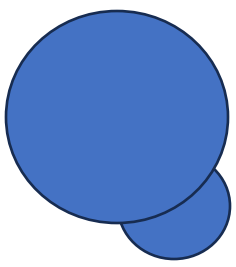
BACHELOR THESIS I,II – GLASS CRUSHER (Year 4)

Objectives

As the team leader for the Software Implementation and motor control team, my objectives for the Glass Crusher project include:

1. Developing and implementing a user interface for real-time system monitoring, feedback, user control inputs and I2C serial communication protocol.
2. My second objective was to develop and implement soft start/stop/pulse operations for smooth running of the crushing motor.
3. My last objective was to perform test procedures in a form of proof-of-concept prototype to ensure the firmware runs smoothly before deployment and integration into the main crusher.

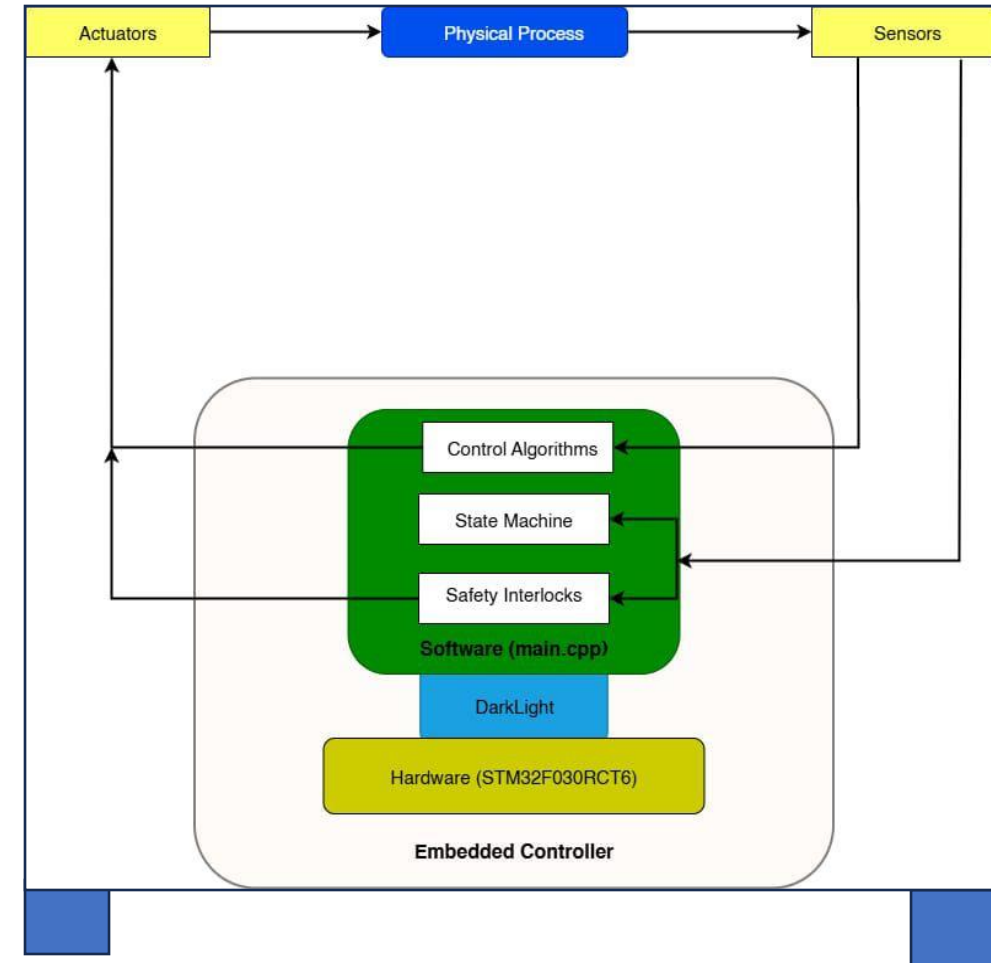


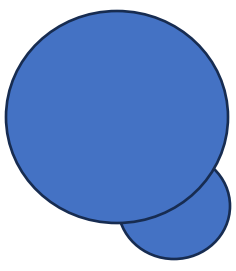


BACHELOR THESIS I,II – GLASS CRUSHER (Year 4)

Role & Contributions

1. Led firmware architecture design and integration of event-triggered/time-triggered real-time control tasks.
2. Developed and tested PID control algorithms with real-time current feedback for crushing motor torque regulation.
3. Designed and implemented safety interlocks including limit switch-based gate control with fault detection and emergency stops.
4. Developed Human-Machine Interface (HMI) with LCD display, LEDs, and push buttons enabling intuitive user control.
5. Conducted unit tests on key firmware modules to ensure robustness and fault tolerance.



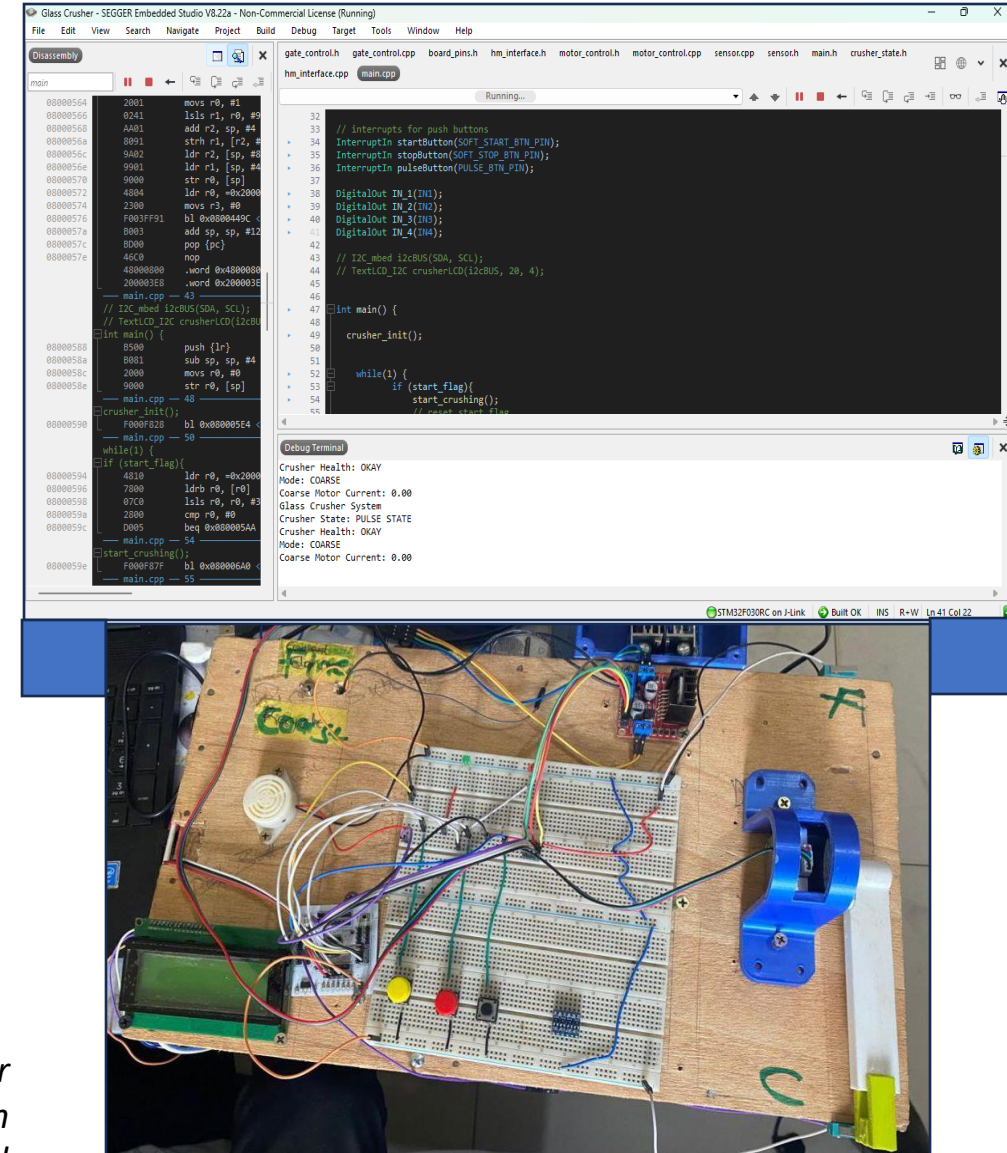


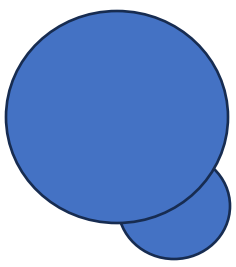
BACHELOR THESIS I,II – GLASS CRUSHER (Year 4)

Technical Highlights

1. Used STM32F030 microcontroller as master controller coordinating motor and gate control with I2C communication.
2. Integrated stepper motor for gate diverter with precision position control, error detection, and visual/audible alerts for fault states.
3. Employed Segger Embedded Studio with DarkLight APIs for efficient embedded development and real-time task scheduling.
4. Firmware implemented soft start/stop, pulse operation modes, and automatic fault response with buzzer and LED indicators.
5. Proof-of-concept validation on a test board demonstrated system reliability under various operational scenarios.

As shown in the images on the right, from the top a screenshot of the Glass Crusher Firmware at the time of development in Segger Embedded Studio. Test board with Nema17 Stepper motor and attached 3D printed bar representing the crushers internal gate which controls the dual output feature of the crusher.





BACHELOR THESIS I,II – GLASS CRUSHER (Year 4)

Outcome & Impact

1. Achieved functional prototype meeting all design objectives for user interface, motor control, and safety features.
2. Provided scalable firmware framework with documented recommendations for RTOS integration, advanced sensor monitoring, and full automation potential.
3. Enabled practical recycling innovation supporting environmental sustainability and local industry empowerment.
4. Set a foundation for future embedded system projects aimed at sustainable industrial automation.

What I Learned

1. Developed strong skills in embedded firmware design by architecting a modular, event/time-driven system and coordinating multiple real-time tasks across user interface modules.
2. Understood the importance of integrated fault detection and safety interlocks, including limit switch position sensing, error signaling, and emergency stop mechanisms.
3. Improved proficiency in hardware-software integration by working closely with microcontroller peripherals, communications protocols (I2C), and embedded development tools.
4. Learned the value of unit testing and systematic verification in embedded systems to catch faults early and ensure reliable, maintainable code.
5. I have also realized the limitations of using an infinite while loop with blocking wait functions and the necessity of an RTOS for a larger system like the crusher

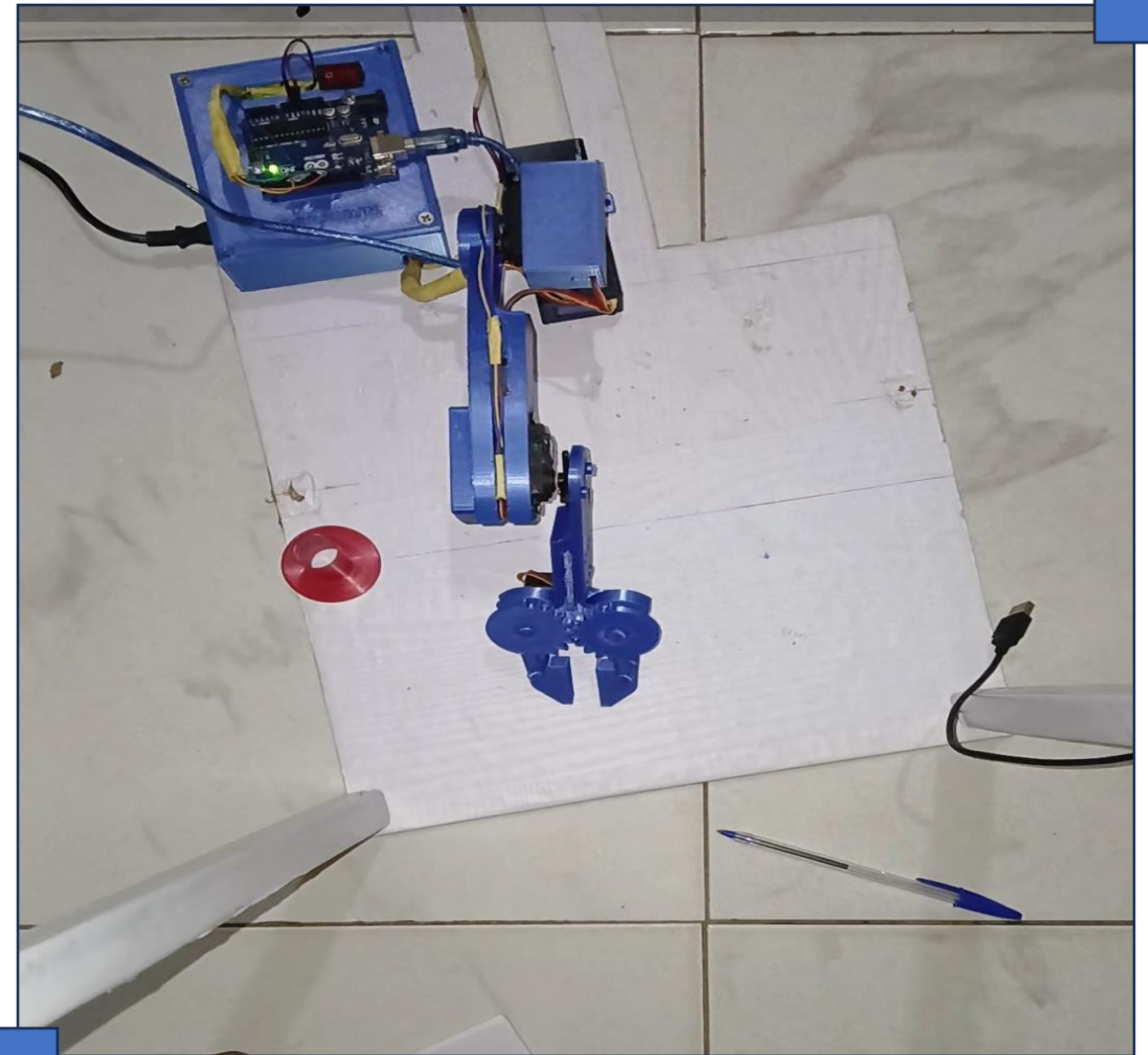
MTR-323 Robotics : Robotron Arm (Year 3)

What is MTR-323 Robotics?

This course introduces path planning, kinematics, dynamics and control in the context of robotic systems. In particular, it focusses on the design aspects of industrial robot manipulators.

Project Overview

As part of the coursework assessment, we were tasked to apply the knowledge obtained from the Robotics module. My team and I based our design on a pick-and-place manipulator with four(4) degrees of freedom in Prismatic-Revolute-Revolute-Revolute(PRRR) configuration because we believe it will provide an optimal balance between our workspace, dexterity, and simplicity, allowing for precise and efficient movement of the objects in a variety of applications such as collection of eggs in a battery cage poultry system, sorting of fruits for processing in industry etc.



MTR-323 Robotics : Robotron Arm (Year 3)

Objectives

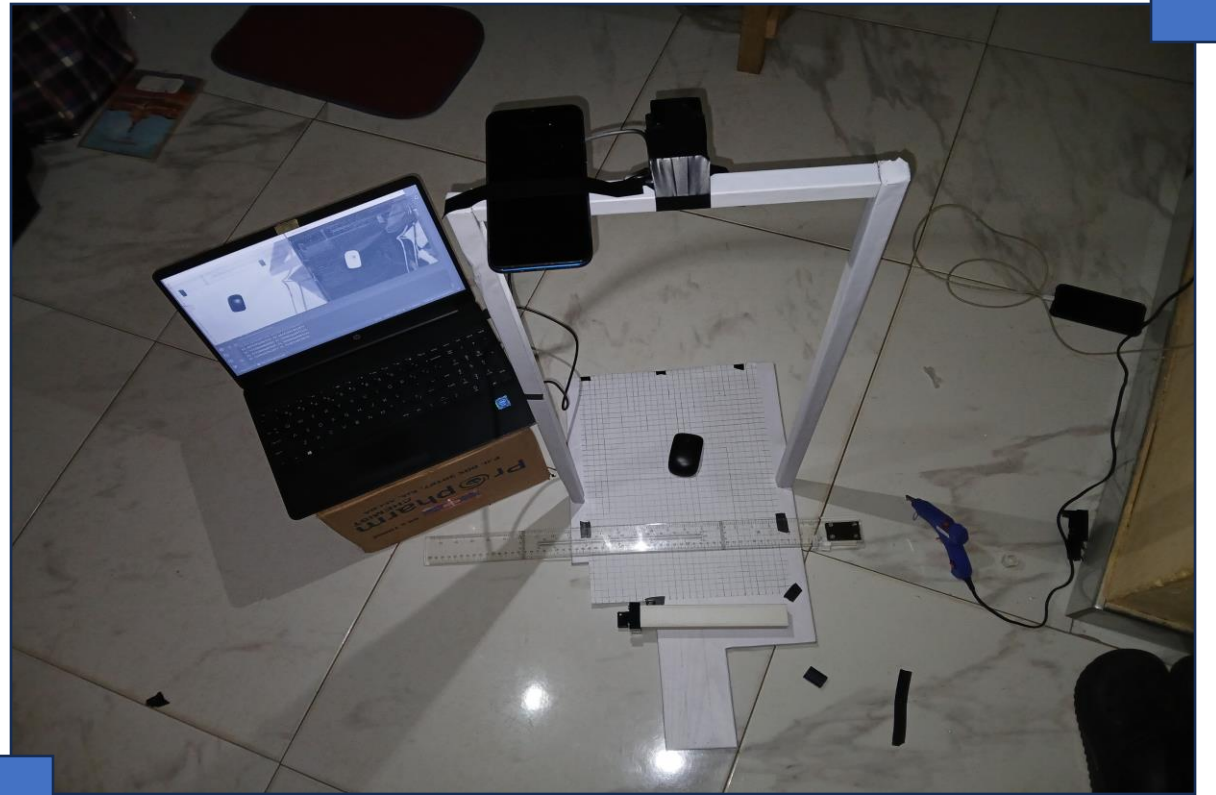
1. To design and manufacture a 4-DoF Robotic arm for pick-and-place tasks.
2. To integrate a webcam as a sensor for robot perception using OpenCV python.



MTR-323 Robotics : Robotron Arm (Year 3)

Role & Contributions

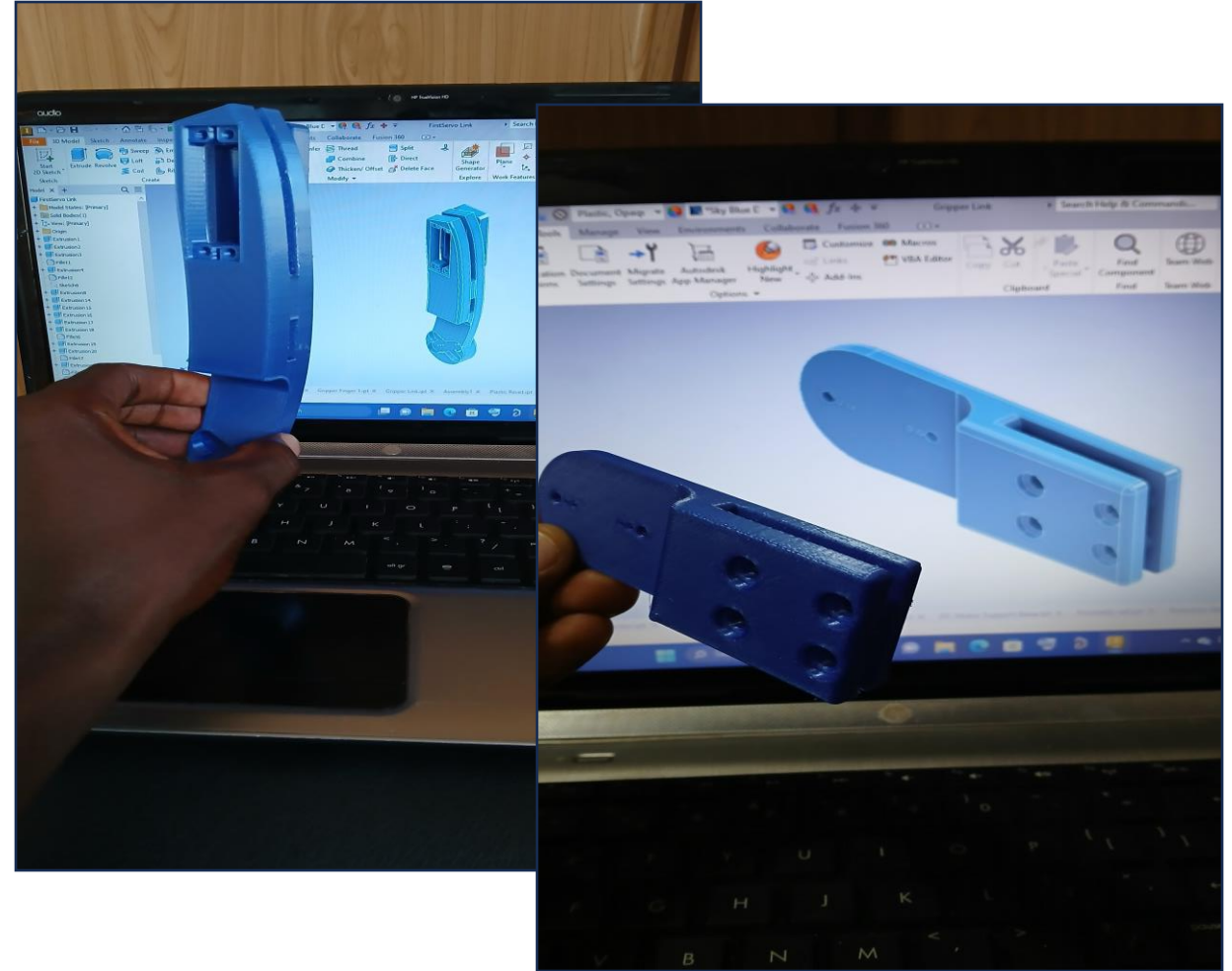
1. Designed and 3D printed the mechanical structure of the robotic arm using Autodesk Inventor Professional.
2. Collaborated with my team to compute the inverse kinematics for the robot arm using link dimensions from the CAD and physical model.
3. Implemented the computed inverse kinematic algorithm on an Arduino Uno R3 using the Arduino IDE.
4. Programmed the webcam using OpenCV's background subtraction libraries to determine the coordinates of the object in 2D – space on the white board used and sent the coordinate data to the Arduino via UART which was used by the inverse kinematic algorithm to precisely move the joint angles to pick and place the object of desire.



MTR-323 Robotics : Robotron Arm (Year 3)

Technical Highlights

1. Refined the mechanical design due to damaged links resulting from haphazard motion of the arm during initial tests.
2. Debugged and refactor both webcam interface code and Arduino code integrating planned path algorithms.
3. Performed additive manufacturing of modelled parts using a Creality Ender V2 – 3D printer.
4. Performed initial motor calibrations to ensure motor angles are within desired range.
5. Tested the arm for days before finally getting it to work as expected without giving up.



MTR-323 Robotics : Robotron Arm (Year 3)

Outcome and Impact

The design and construction of the 4-DoF Robotic arm shaped my understanding of robotic manipulators in general and lay a foundation for future works.

My Robotics course instructor, Mr.. Obed Doku, second from left in the image was pleased with performance in applying the theory learnt from class which got my team and I an A in the Robotics coursework assessment. This proves that we were able to achieve our project objectives from beginning to finish.

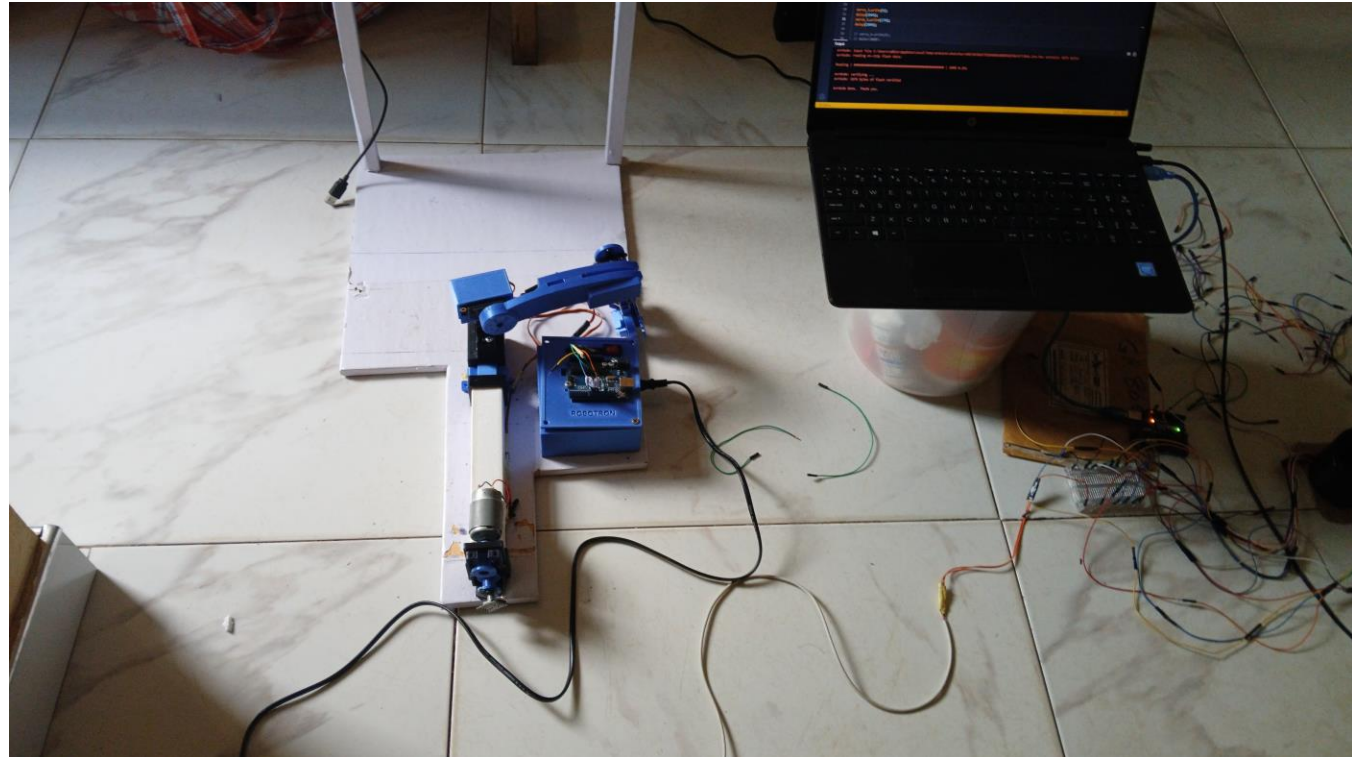


MTR-323 Robotics : Robotron Arm (Year 3)

What I have Learned

The Robotron project taught me valuable lessons which I apply to projects I work on ever since. The first lesson reminds me that getting a robot to work as expected requires utmost dedication, time, effort and a tenacious spirit. The robot can only work when you never give up.

Second lesson more of technical, after damaging parts of the first printed robot arm, I realized I focused on aesthetics instead of functionality, this has shaped my understanding of hardware design. Another lesson I learnt is about the inverse proportion dependence of the software and the hardware on each other. For a bad hardware design, the software needs to compensate for extra functionalities in order to get a robust and a reliable system. I realized this when I smoothen the motion of the arm to prevent jerky movements in order not to damage the poorly designed joints and links.



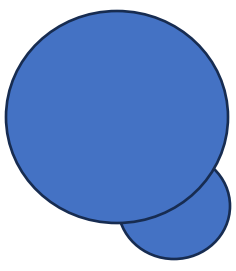
Other Academic Projects

Project Lead : Digital Clock Prototype – Year Two (2)

This digital clock was developed as a core assessment for the Product Development Course, demonstrating the full lifecycle of creating an embedded system. Leading a team of five mechatronics students, I coordinated the firmware development using the Mbed OS framework—specifically leveraging Timer and Ticker APIs—to program an NXP KL25Z development board powered by an ARM Cortex-M0+ processor. My role encompassed end-to-end ownership, from initial code architecture and task delegation to hands-on PCB soldering, system debugging, and iterative testing. We successfully transformed the initial concept into a fully functional product prototype, navigating challenges such as electrical faults and firmware timing issues. This project served as a practical application of embedded programming, hardware integration, and agile problem-solving in a team environment.



The images above shows visuals from the project. Starting from left in the first image above shows myself (wearing a white T-shirt in the middle) leading my team to troubleshoot an electrical fault in the course of soldering. Second image shows myself doing testing and re-soldering work. The last two (2) images shows the completed prototype of the Digital Clock.



Other Academic Projects

Mbed Application Programming Interface (API) – Year 4

The retirement of the Mbed OS platform in 2024 created a critical gap in our embedded systems curriculum at Koforidua Technical University. The subsequent move to low-level STM32 programming (bare-metal, CMSIS, HAL) proved excessively difficult for sophomore engineering students, hindering their ability to engage with core course material effectively.

I led a team of top students in the Mechatronics department to design and develop an educational STM32 API, inspired by the retired Mbed framework, to abstract low-level hardware complexity for sophomore engineering students. I managed the full project lifecycle using Git/GitHub, delivering a tool that allowed students to prioritize application development over register configuration. The solution was later adopted by university staff, scaled nationally under the name "DarkLight," and continues to aid embedded systems education across Ghana.

```
main.cpp  DigitalOut.h  main.h  DigitalOut.cpp  DigitalIn.cpp  DigitalIn.h  Timer.cpp  PWM.cpp  InterruptIn.cpp

1  /**
2   *      MBED API PROJECT
3   *
4   *      *****
5   *
6   *      ----- END-OF-HEADER -----
7   *
8   Drivers : Low Level Drivers, CMSIS_5, STM32F0XX
9   Language : C/C++
10  Tools   : Embedded studio
11  Target Device : STM32F030RCT6
12  Collaborators : Mr. Richard Mensah, Edwin Setsoafia, Enos Essandoh,
13                  Amidu Lukman, Lawson Chinedu, Paul
14
15  Comment Style: @brief describes in brief the functionality
16                  @param description of parameters
17                  @important needs to be critically examined
18                  // For comments that are not lengthy
19
20  NB: This is a standard for code clarity and better documentation
21  Adhere to it!
22
23  */
24
25
26
```

Internship Project (Industrial Attachement)

Fortress E&P Ghana – Year 3

My internship at Fortress Ghana was a highly rewarding experience that allowed me to develop practical skills in web development, robotics, mechanical design, and teamwork through diverse and challenging projects. Despite initial challenges due to the absence of key personnel, the supportive workplace environment and collaborative efforts of the team ensured I could overcome these hurdles and contribute meaningfully to the projects.

The internship provided invaluable hands-on exposure, bridging the gap between academic knowledge and industry practice. Overall, the experience has significantly enhanced my technical abilities, problem-solving skills, and confidence in handling multidisciplinary engineering tasks.

This internship was highly relevant to my studies in Mechatronics Engineering and provided valuable practical exposure to robotics and systems programming. By working on projects such as enhancing Python programming skills and contributing to a drone prototype, I gained hands-on experience and deepened my understanding of systems integration and robotics applications. This experience has prepared me to apply these skills in both academic and professional settings.



Internship Project (Industrial Attachement)

Fortress E&P Ghana – Activities Performed

Welcome Guest!

Full Name*
Edwin Setsoafia

Personal Contact*
0544897559

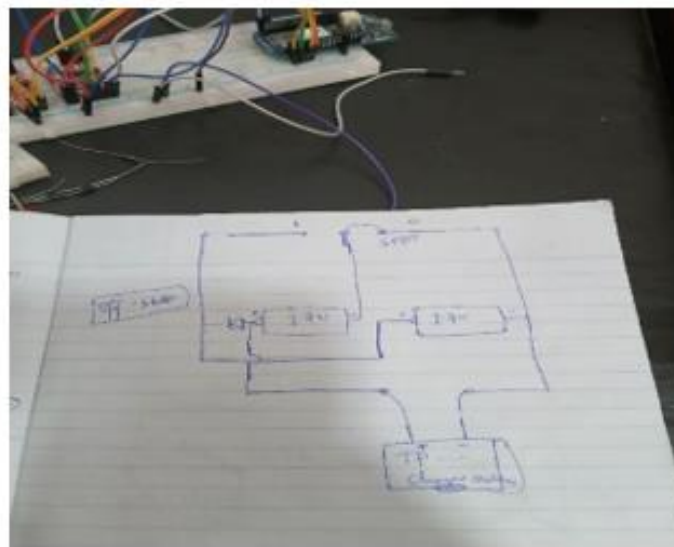
Emergency Contact*
0243165809

Company*
esK Robotics

Retake Photo **Submit**

Generate Report

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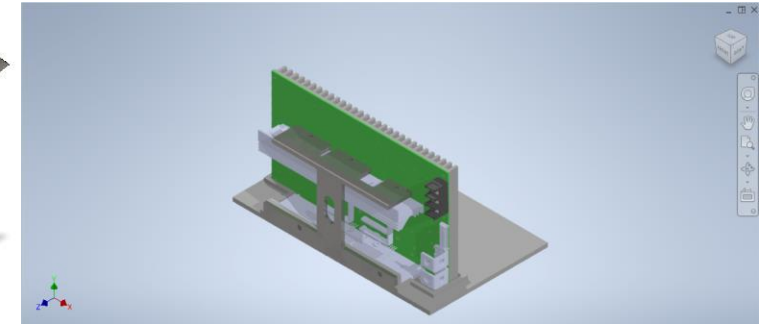
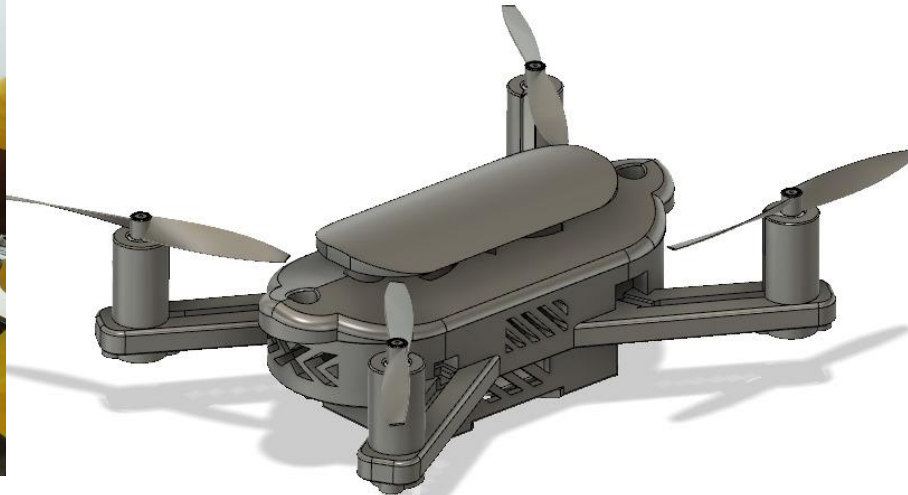


Tasks Performed

1. Assisted in the development of a Visitor Log System for use in Fortres Ghana.
2. Assisted in the design and testing of a battery charger circuit.
3. Assisted in the design of a mini-drone using Fusion 360.
4. Assisted in the design of a mechanical mount for use in an R&D ROV.
5. Assisted in the testing of ROV components like camera and sensors.

From left to right, a screenshot showing visitor details being taken on the frontend, a report showing all records of visitors in the database, and below the visitor record; the battery charger circuit implemented on the breadboard using TP4056 charger module.

Internship Project (Industrial Attachement) Fortress E&P Ghana – Activities Performed

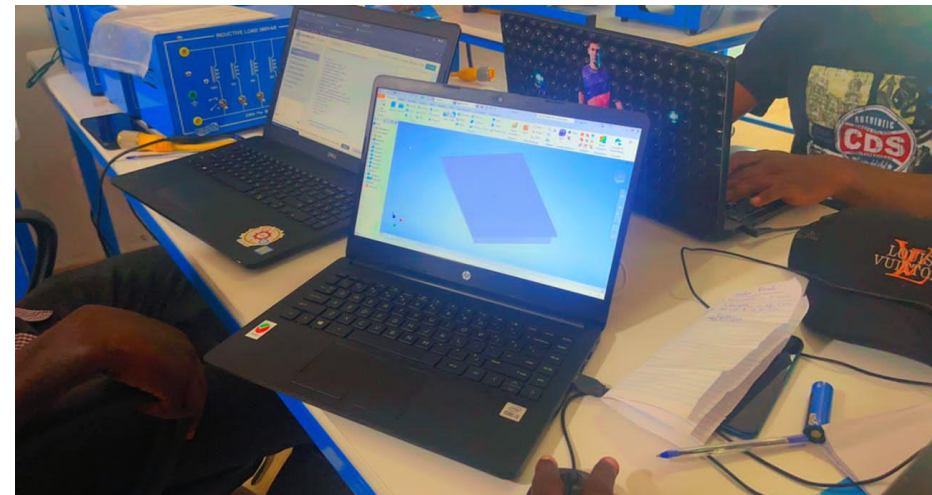


From left to right, a screenshot showing the testing of KUV sensors connected to a server using fibre-optic cables, a drone designed in Fusion 360 and finally, a mechanical mount for a printed circuit board in an ROV compartment.

Extracurricular Projects

Campus Robotics CAD Team Lead – Year 2

During my second year, I served as the CAD team lead for my Campus Robotics Club, supervising the mechanical design of a smart bin system that was showcased at an engineering fair at Accra Technical University. I designed and 3D-printed a battery casing for the bin, as well as an ultrasonic sensor holder. The accompanying images show me, wearing a green shirt and facing the camera, with my CAD teammates. Another image features the electrical and software teams of the Campus Robotics Club.



Extracurricular Projects

STEM Coach: October 2025



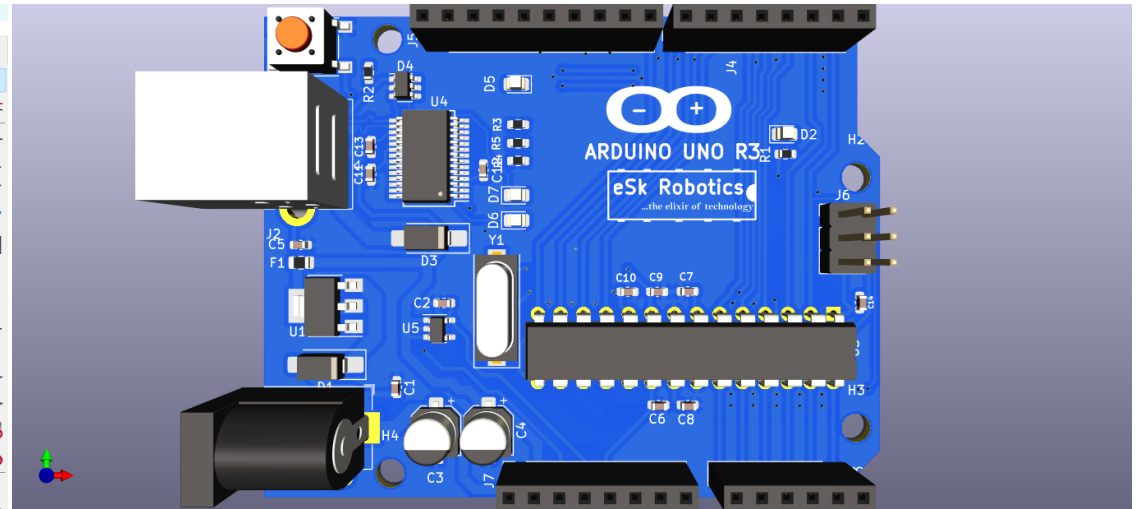
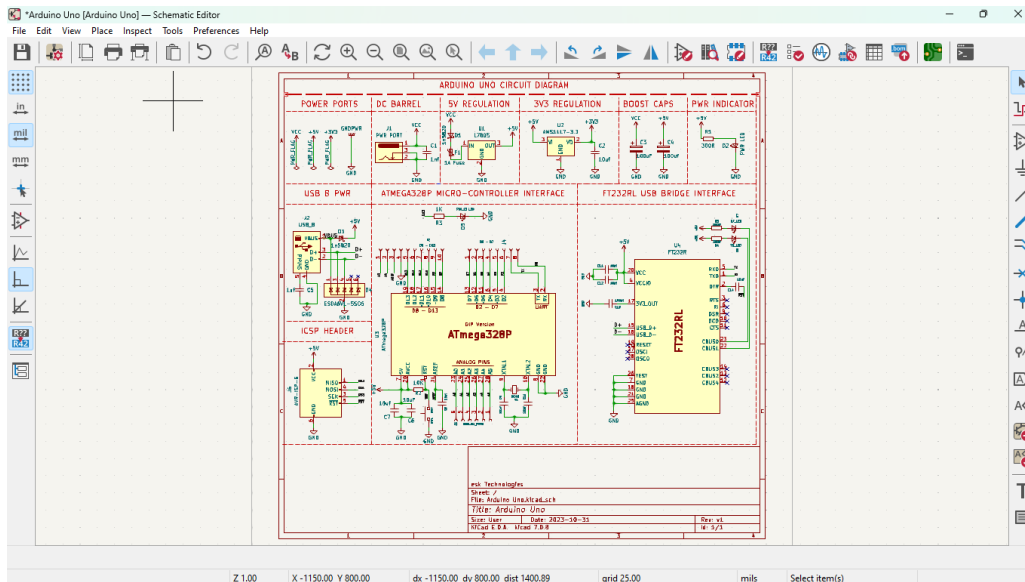
Assisted in mentoring students from Koforidua Secondary Technical in the development of a smart dispenser that employs voice activation and hand-motion gestures to dispense water to users. This innovative device was showcased at the Innovate Koforidua Challenge, organized by Eastern Tech Hub.



Personal Projects

Custom Arduino Uno R3

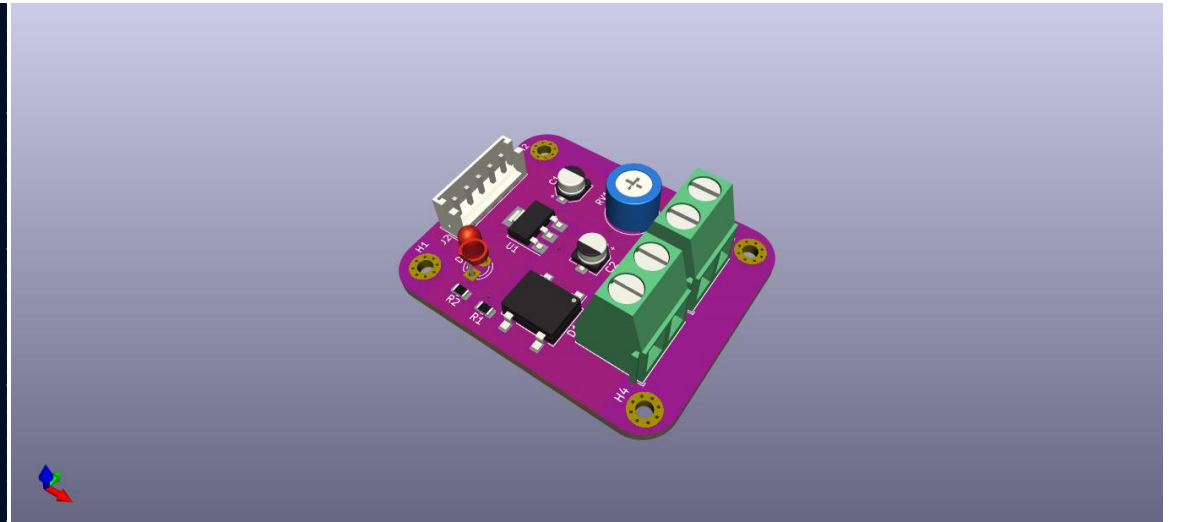
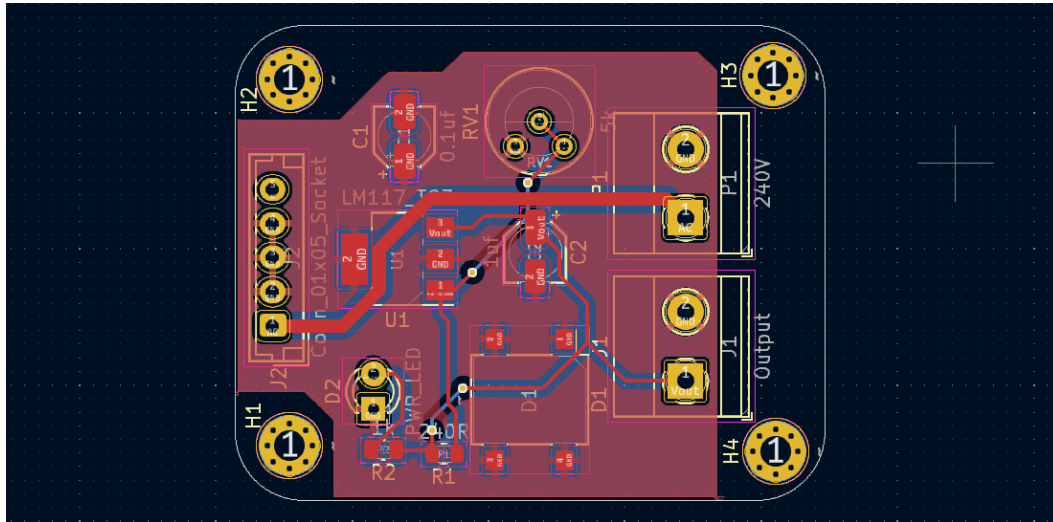
Designed a clone of the Arduino Uno R3 in KiCAD EDA, applying skills in PCB schematic capture, component placement, and effective power and signal routing. Although there are no plans to fabricate the PCB, as my works primarily involves STM32 microcontrollers, this project significantly strengthened the PCB design skills acquired during my Product Development coursework using Altium EDA.

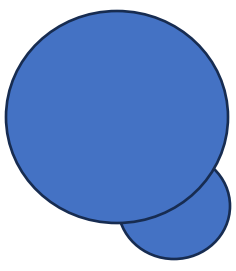


Personal Projects

5V Linear Power Supply

Through my Analogue Electronics I coursework, I learned to design and simulate linear power supplies, essential components in electronic systems. During the course, I used Multisim, a circuit simulation tool, to model the performance of these circuits. After completing the Product Development course, I revisited this design and recreated it in KiCAD. The image below displays the PCB layout and routed schematic in KiCAD, which is yet to be fabricated to power my upcoming robotic hand manipulator project.

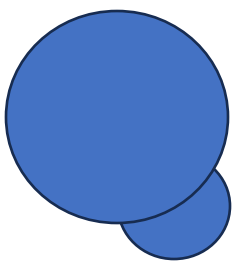




Personal Projects

Self-Balancing Robot Simulation (MATLAB)

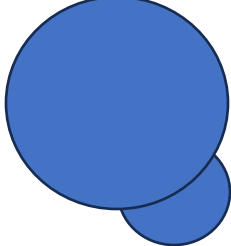
The Control System I and II courses laid the foundation for my understanding of control theory. During these courses, I used MATLAB to simulate the dynamics of an inverted pendulum system. To further deepen my understanding of the concepts, I performed dynamic simulations using the Simscape multi-body library, where I applied advanced modeling techniques to analyze system behavior and control responses efficiently.



My Thoughts On Robotic Manipulation Research

I am currently in the conceptual stage of designing a Robotic Hand Manipulator. Ongoing research in this field has made noteworthy progress, and I am eager to contribute further. When comparing the currently available hardware to the human hand, there are significant differences. I strongly believe that, this disparity is a key reason why substantial breakthroughs in grasping and manipulation remain elusive. To advance further, a separation of concerns is necessary in the hardware design of manipulators that aim to mimic the human hand. Achieving dexterous manipulation requires designing hardware tailored to specific use cases. For example, our biological hands are not suited for handling hazardous substances, which justifies the use of metallic robotic hands for dangerous tasks. However, employing the same metallic hand for folding clothes or picking a single plate from a stack is inefficient—akin to using a human skeletal arm to grasp a mug. Are we using the right tools for the wrong tasks? By specializing robotic manipulators for distinct applications, rather than generalizing them, significant progress can be made in this research domain. For industrial tasks, manipulators can be designed to adapt to the shape of objects, optimizing the manipulation process. For tasks such as filling a coffee cup, newly developed compact and flexible materials—combined with under-actuation—can help achieve fluid-like manipulation, similar to the function of a soft human palm.

In summary, to manipulate everyday objects like the human hand, both physical structure and functionality should closely resemble the human hand. For tasks beyond human capability, the manipulator's structure should be changed while maintaining human-like behavioral algorithms.



Completed Certifications



Ongoing Certifications/Courses

1. Zero-To-Mastery: Deep Learning with Pytorch
2. Robotic Manipulation: Perception, Planning and Control, Russ Tedrake (MIT)